

SQL Pivot Operation



Pivot Operation

- ```
select *
from sales
pivot (
 sum(quantity)
 for color in ('dark','pastel','white')
)
order by item name;
```

| item_name | clothes_size | dark | pastel | white |
|-----------|--------------|------|--------|-------|
| dress     | small        | 2    | 4      | 2     |
| dress     | medium       | 6    | 3      | 3     |
| dress     | large        | 12   | 3      | 0     |
| pants     | small        | 14   | 1      | 3     |
| pants     | medium       | 6    | 0      | 0     |
| pants     | large        | 0    | 1      | 2     |
| shirt     | small        | 2    | 4      | 17    |
| shirt     | medium       | 6    | 1      | 1     |
| shirt     | large        | 6    | 2      | 10    |
| skirt     | small        | 2    | 11     | 2     |
| skirt     | medium       | 5    | 9      | 5     |
| skirt     | large        | 1    | 15     | 3     |

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explain this

This slide explains the **Pivot Operation** in SQL, which is used to transform rows into columns, making it easier to analyze data.

## SQL Syntax in the Slide

```
sql

select *
from sales
pivot (
 sum(quantity)
 for color in ('dark', 'pastel', 'white')
)
order by item_name;
```

### ✓ What it does:

- The query is run on a table called `sales`.
- `pivot (...)` is used to convert **values in the** `color` **column** (`'dark'`, `'pastel'`, `'white'`) into **columns**.
- For each color, it **sums the** `quantity` of sales.
- The result is grouped by the other columns (like `item_name` and `clothes_size`).

- Finally, it orders the output by `item_name`.



## Output Table Explanation

| item_name | clothes_size | dark | pastel | white |
|-----------|--------------|------|--------|-------|
| dress     | small        | 2    | 4      | 2     |
| dress     | medium       | 6    | 3      | 3     |
| dress     | large        | 12   | 3      | 0     |
| pants     | small        | 14   | 1      | 3     |
| pants     | medium       | 6    | 0      | 0     |
| pants     | large        | 0    | 1      | 2     |
| shirt     | small        | 2    | 4      | 17    |
| shirt     | medium       | 6    | 1      | 1     |
| shirt     | large        | 6    | 2      | 10    |
| skirt     | small        | 2    | 11     | 2     |
| skirt     | medium       | 5    | 9      | 5     |
| skirt     | large        | 1    | 15     | 3     |

Each row represents a unique `(item_name, clothes_size)` combination, and the `dark`, `pastel`, and `white` columns show the **total quantity sold** for that color.



## Why Use Pivot?

- Makes reports more readable.

- Easier to compare color-wise sales for each item and size.
- Useful for business analytics.

Let me know if you'd like to see how this would look in standard SQL (like PostgreSQL or MySQL), or how to unpivot it!

explain pivoting in OLAP

Great question!



## Pivoting in OLAP (Online Analytical Processing)

In the context of **OLAP**, **pivoting** is a powerful operation used to **reorganize** and **analyze** multi-dimensional data from different perspectives.

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## What is Pivoting?

**Pivoting** refers to **rotating the data axes** (dimensions) of a data cube to get a new perspective on the data.

Think of it like rotating a Rubik's cube to look at a different face — same data, different view.

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## In OLAP Terms:

OLAP organizes data into **cubes** with dimensions (e.g., time, product, region) and **measures** (e.g., sales, profit).

**Pivoting lets you:**

- Move dimensions between rows and columns
  - Change the focus of analysis without modifying the underlying data
  - Explore different summaries interactively
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## Example:

Imagine you have a data cube with:

- **Dimensions:** Time, Product, Region
- **Measure:** Sales

**Pivot View 1:**

| Region | Jan | Feb | Mar |
|--------|-----|-----|-----|
| USA    | 100 | 120 | 130 |
| UK     | 90  | 110 | 115 |

**Pivot View 2 (after pivoting Time and Region):**

| Month | USA | UK  |
|-------|-----|-----|
| Jan   | 100 | 90  |
| Feb   | 120 | 110 |
| Mar   | 130 | 115 |

The data is the same, but we've pivoted the cube to focus on **months in rows** and **regions in columns**.

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## Why Use Pivoting in OLAP?

- **Flexible analysis:** See your data from any angle.
  - **Faster insights:** Quickly switch views without rewriting queries.
  - **Interactive tools:** OLAP tools (like Microsoft Excel, Tableau, or Oracle OLAP) let users pivot data visually.
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## Summary

| Feature      | Description                                                         |
|--------------|---------------------------------------------------------------------|
| Operation    | Re-orientes dimensions                                              |
| Used in      | OLAP cubes (multidimensional data)                                  |
| Goal         | Gain different perspectives without changing data                   |
| Common tools | Excel, Tableau, Power BI, Oracle OLAP, SQL Server Analysis Services |

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Would you like to see how pivoting compares to other OLAP operations like **drill-down**, **roll-up**, or **slice/dice**?

Do you like this personality